

5. Linear programming		
5.1	Formulation of problems as linear programs.	
5.2	Graphical solution of two variable problems using ruler and vertex methods.	
5.3	Consideration of problems where solutions must have integer values.	

Glossary for D1

1. Algorithms

In a list containing N items the 'middle' item has position $\lceil \frac{1}{2}(N+1) \rceil$ if N is odd, $\lceil \frac{1}{2}(N+2) \rceil$ if N is even, so that if $N=9$, the middle item is the 5th and if $N=6$ it is the 4th.

2. Algorithms on graphs

A **graph** G consists of points (**vertices** or **nodes**) which are connected by lines (**edges** or **arcs**).

A **subgraph** of G is a graph, each of whose vertices belongs to G and each of whose edges belongs to G .

If a graph has a number associated with each edge (usually called its **weight**) then the graph is called a **weighted graph** or **network**.

The **degree** or **valency** of a vertex is the number of edges incident to it. A vertex is **odd (even)** if it has **odd (even)** degree.

A **path** is a finite sequence of edges, such that the end vertex of one edge in the sequence is the start vertex of the next, and in which no vertex appears more than once.

A **cycle (circuit)** is a closed path, i.e. the end vertex of the last edge is the start vertex of the first edge.

Two vertices are **connected** if there is a path between them. A graph is **connected** if all its vertices are connected.

If the edges of a graph have a direction associated with them they are known as **directed edges** and the graph is known as a **digraph**.

A **tree** is a connected graph with no cycles.

A **spanning tree** of a graph G is a subgraph which includes all the vertices of G and is also a tree.

A **minimum spanning tree (MST)** is a spanning tree such that the total length of its arcs is as small as possible. (MST is sometimes called a **minimum connector**.)

A graph in which each of the n vertices is connected to every other vertex is called a **complete graph**.

The **travelling salesman problem** is 'find a route of minimum length which visits every vertex in an undirected network'. In the '**classical**' problem, each vertex is visited once only. In the '**practical**' problem, a vertex may be revisited.

For three vertices A , B and C , the **triangular inequality** is 'length $AB \leq \text{length } AC + \text{length } CB$ ', where AB is a longest length.

A **walk** in a network is a finite sequence of edges such that the end vertex of one edge is the start vertex of the next.

A walk which visits every vertex, returning to its starting vertex, is called a **tour**.

3. Critical path analysis

The **total float** $F(i, j)$ of activity (i, j) is defined to be $F(i, j) = l_j - e_i - \text{duration}(i, j)$, where e_i is the earliest time for event i and l_j is the latest time for event j .